Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/18/2019
PI Name:	Sevick-Muraca, Eva Ph.D.		
Project Title:	Quantification of the Lymphatic Pump Strength and Assessment of CSF Drainage Into the Lymphatics during HDT		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Eva.sevick@uth.tmc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-500-3560
Organization Name:	The University of Texas Health Science Center		
PI Address 1:	1825 Pressler Street, SRB 330A		
PI Address 2:	The Brown Foundation Institute of Molecular Medicine		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	05/01/2017	**End Date:**	07/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2018 (original end date was 4/30/2018) per E. Urquieta/TRISH (Ed., 5/17/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rasmussen, John C. Ph.D. (The University of Texas Health Science Center)		
Grant/Contract No.:	NNX16AO69A-FIP0001		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Focused Investigation Project To build infrastructure for monitoring cerebral spinal fluid/interstitial fluid (CSF/ISF) outflow, thoracic/cervical congestion, and peripheral lymphatic pumping in human subjects. Specifically, the project team will build a next generation near-infrared fluorescence camera system for lymphatic imaging that will enable more rapid and more sensitive imaging than was possible with prior near-infrared fluorescence lymphatic imaging (NIRFLI) prototypes. The team will also develop and execute protocols for assessing the peripheral lymphatic "pump," internal and external lymphatic flow from the head and neck, and cervical lymphatic congestion during head down tilt (HDT) in humans.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This one year project demonstrated: - intrathecal administration of indocyanine green (ICG) results in drainage through the subarachnoid space to the cervical lymph nodes in swine and from the spinal canal directly to the peripheral lymphatics. Due to the comparatively deep spinal processes in swine as compared to humans, drainage could not be visualized without sacrifice. - mucosal administration of ICG in the palantine tonsils of human subjects enabled visualization of drainage into the cervical lymph nodes, potentially providing a diagnostic for measuring cerebrospinal fluid (CSF) outflow in Spaceflight Associated Neuro-ocular Syndrome (SANS). In addition, this project demonstrated near-infrared, IsCMOS (Intensified CMOS Camera) system for evaluating the role of lymphatic function in cephalad fluid shifts.
Task Progress:	Phase I: MONTHS 0-6 (1) Build a near-infrared fluorescence lymphatic imaging (NIRFLI) system for deployment in human and swine studies with future capacities for battery operation necessary for terrestrial microgravity simulations of human physiology; a. The technology will replace a 16-bit CCD chip with a 16-bit sCMOS chip for faster data read-out rates that will enable greater sensitivity, speed up image acquisition, allow ambient light rejection for real-time display, and video-rate capture of lymphatic pumping. b. The technology will include proprietary military grade, Gen III intensification to amplify and gate-out ambient light in a manner that our prior CCD based systems which operated at slow acquisition speeds could not enable. c. The technology will be uniquely adapted with auto-focus capabilities sufficient for capturing wide field views of lower extremities as well as smaller fields of view of cervical regions. The technological developments will be necessary to insure that internal lymphatics and CSF/ISF drainage from intrathecal ICG injections can be sensitively imaged in large animal models and possibly in humans. (2) Develop a clinical protocol that measures lower extremity lymph pump rate as well as internal and external cervical lymphatic drainage as a function of position (supine, increasing HDT--head down tilt, decreasing HDT). In addition, these parameters may be assessed pre- and post- Valsalva maneuver, hyperventilation, breath holding, and head turning right and left to investigate the potential role of internal jugular vein compression. a. We will design and build a custom 32 gauge serializable spinal tap needle with delimiter for mucosal injections in the area of the palatine tonsils that will enable ICG imaging of the internal cervical lymphatics and internal lymphatics in human studies. b. University of Texas (UT) Health Institutional Review Board (IRB) approvals will be gained. c. FDA (Food & Drug Administration) amendment will be made. Phase II: MONTHS 6-12 (1) Install the completed portable NIRFLI system, perform and test ambient light rejection performance at UTHealth and its veterinary Center for Laboratory Animal Medicine and Care (CLAMC) facilities for human and swine studies, respectively; (2) Perform a site initiation visit with participating clinicians; (3) Consent and conduct human studies on five normal adults recruited in conjunction with clinical collaborators; and (4) Conduct animal studies on up to five swine for developing intrathecal ICG injection protocols to image CSF transfer to the cervical lymphatics. Deliverables: MONTH 12 (1) Reduction to practice of technical advances developed in our laboratory in order to produce a near-infrared fluorescence lymphatic imaging system capable of faster, more sensitive measurements in the presence of ambient light and with autofocusing capabilities; (2) Peer review publication on imaged lymphatic fluid shifts, active pumping lymphatic transport, and internal cervical lymphatic flow in humans in response to HDT; and Technical Standard Operating Procedures (SOPs) and manuscript for assessing CSF drainage into lymphatics following non-surgical intrathecal administration of ICG in a large animal model with intent to translate into non-surgical testing in humans and non-human primates. In addition, we partnered with Rutgers' Mark Pierce in a synergy grant to test whether short-wave near-infrared (SWIR) fluorescence can provide a better imaging result than NIR (near-infrared) imaging using either a SWIR excitable nanoparticle or ICG. We tested SWIR (using loaner equipment) versus our NIR equipment in swine. (2) Key findings. -Due to the comparatively deep spinal processes of swine as compared to humans, imaging CSF outflow into cervical lymph nodes after intrathecal injection of ICG or short-wave near-infrared (SWIR) is not possible in living swine. -Using NIR fluorescence imaging we find that intrathecal injection of swine resulted in the filling of subarachnoid space and only upon sacrifice, found drainage from the subarachnoid space to the cervical lymph nodes. We also found drainage from the spinal canal to lymph nodes in the hind quarters as we have previously reported in mice. This result is exciting as it confirms direct CSF communication with the peripheral lymphatic system. -Because the intrathecal route of administration of ICG has never been performed in humans, toxicity and safety testing is required. Nonetheless, we found that injection of ICG into the palantine tonsils of normal human subjects enabled visualization of the same lymphatic drainage route into the cervical lymph nodes as CSF outflow occurs. Thus, we have developed a potential diagnostic for CSF outflow. -SWIR signatures were not detectable using SWIR particles. SWIR imaging of ICG was not superior to NIR imaging of ICG. -Lymphatic pumping function in the legs and drainage from the head profoundly varies with HDT. (3) Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal. NIR imaging using the developed IsCMOS system provides a robust methodology to image lymphatic function in humans. NIR imaging provides a new diagnostic to evaluate CSF outflow which could provide an important measure for SANS as well as neuroinflammation. Contrary to published results by others, SWIR imaging provides less contrast than NIR imaging owing to water absorption and impaired penetration depths. Near-infrared imaging after intrathecal injection of ICG could provide a better assessment of CSF outflow in humans. Future safety and toxicity studies are needed to enable this route of administration. (4) Results. HDT data suggests that the lymphatics play an important role in the etiology of SANS.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Articles in Peer-reviewed Journals	Rasmussen JC, Kwon S, Pinal A, Bareis A, Velasquez FC, Janssen CF, Morrow JR, Fife CE, Karni RJ, Sevick-Muraca EM. "Assessing lymphatic route of CSF outflow and peripheral lymphatic contractile activity during head-down tilt using near-infrared fluorescence imaging." Physiol Rep. 2020 Feb;8(4):e14375. https://doi.org/10.14814/phy2.14375 ; PMID: 32097544; PMCID: PMC7058174 , Feb-2020

Articles in Peer-reviewed Journals	Zhu B, Kwon S, Rasmussen JC, Litorja M, Sevick-Muraca EM. "Comparison of NIR versus SWIR fluorescence image device performance using working standards calibrated with SI units. " IEEE Trans Med Imaging. 2020 Apr;39(4):944-51. https://doi.org/10.1109/TMI.2019.2937760 ; PMID: 31478842 , Apr-2020